Spikingformer+Spik4lite Demo

Introduction

This is a demo implementation of "**Spik4lite: Refactoring Neuromorphic Sparsity for Efficient Spiking Neural Networks on Commodity Edge Devices**". The demo follows the **train-and-infer** pipeline to show the basic usage of Spikingformer+Spik4lite.

Directory

The `Spikingformer` is the root and its structure is described as below.

```
Spikingformer/
├──energy_consumption_calculation/
├──README.md
    ├──cifar10
    │    ├──cifar10.yml
    │    ├──model.py
    │    ├──SOPs_consumption_on_cifar10.py
    │    ├──Spikingformer Power profile.csv
    │    ├──Spikingformer+Spik4lite  Power profile.csv
    │    ├──test.py
    │    ├──train.log
    │    ├──train.py
    ├──cifar100
    │    ├──cifar100.yml
    │    ├──model.py
    │    ├──SOPs_consumption_on_cifar100.py
    │    ├──test.py
    │    ├──train.py
    ├──cifar10-dvs
    │    ├──autoaugment.py
    │    ├──model.py
    │    ├──SOPs_consumption_on_CIFAR10DVS.py
    │    ├──test.py
    │    ├──train.py
    │    ├──utils.py
    ├──dvs128-gesture
    │    ├──autoaugment.py
    │    ├──model.py
    │    ├──SOPs_consumption_on_DVS128Gesture.py
    │    ├──test.py
    │    ├──train.log
    │    ├──train.py
    │    ├──utils.py
```

* `README.md` provides the general overview and introduction of the project, as well as instructions on how to run the code.

Running the Demo

The whole pipeline contains three major steps: (1) activate the Conda Environment, (2) Run the train code and Run the SOPs consumption code.

(1) activate the Conda Environment

```
cd Spikingformer

conda activate Spik4lite
```

(2) Run the code

Runing on CIFAR10

Setting hyper-parameters in cifar10.yml

```
cd cifar10
python train.py
```

Runing on CIFAR100

Setting hyper-parameters in cifar100.yml

```
cd cifar100
python train.py
```

Runing on DVS128 Gesture

```
cd dvs128-gesture
python train.py
```

Runing on CIFAR10-DVS

```
cd cifar10-dvs
python train.py
```

Runing on SOPs consumption

```
python SOPs_consumption_on_cifar10.py
```